

DIRECTIONS *for questions 1 to 4:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

In 1932, Robbins redefined the field of 'Economics' as essentially a series of relationships. To distance himself from "psychological hedonism" of wants and desires, he explicitly used Max Weber's analysis of means–ends rationality: "Economics . . . is concerned with that aspect of behavior which arises from the scarcity of means to achieve given ends" (Robbins 1932: 24; cf. Maas 2009: 511). Nonetheless, the foundation of his anti-hedonistic argument is unreservedly Paretian: "All that we need to assume as economists is the obvious fact that different possibilities offer different incentives, and that these incentives can be arranged in order of their intensity". The idea of ordering preferences is derived directly from the conception of scarcity of means, assuming [that] for each individual, goods can be ranged in order of their significance for conduct; and that, in the sense it will be preferred, we can say that one use of a good is more important than another.

Following Pareto, he believes that [...] individuals can arrange their preferences in an order, and in fact they do so. As he explains, individuals do make rational disposal of goods in a consistent manner, but their respective rankings are only a matter of normative valuation. Robbins recognizes that individual choices are not always consistent, since exchange, production, fluctuation – all take place in a world in which people do not know the full implications of what they are doing. This is why it is assumed that people act purposively, in a perfectly rational manner, by knowing everything they need to know in order to take the right decision. These assumptions are made to enable us to study, in isolation, tendencies which, in the world of reality, operate only in conjunction with many others.

Yet, the justification of the assumption of rationality differs significantly from the one that was made in classical political economy. It is no more the result of a long and accurate observation, an empirical generalization established by the science of psychology, but an obvious and self-evident proposition based on introspection in the pure marginalist tradition.

Even so, he does not endorse Pareto's definitive divorce from psychology either, recognizing that the very structure of means–ends rationality is "psychical not physical". Let us explain why there is no discrepancy here: challenging Pareto's views, he affirms that "it is really not possible to understand the concepts of choice, of the relationship of means and ends, the central concepts of our science, in terms of external data". The assumption of completely rational conduct is after all a psychological assumption showing how people behave in situations of scarcity. The problem is that psychology has failed, according to Robbins, to measure empirically "the scales of relative valuation" of individual feelings. We can only feel an individual's subjective valuation process without being able to observe it as a fact. Economists should therefore abandon this Sisyphean task which finally leaves the door open to subjective, i.e. normative judgments. Hence, Robbins minimized the importance of psychology to the simple and indisputable facts of experience in order to understand economizing behaviour under scarcity.

Q1. The 'Sisyphean task' mentioned in the last sentence of the passage refers to:

- ☐ a) the objectification of individual feelings.
- ☐ b) defining the process of individuals' subjective valuations.
- ☒ c) **empirical measurement of individual feelings.** ✖ Your answer is incorrect
- ☐ d) identifying an individual's subjective valuation process.

Q2. Which of the following best captures Max Weber's analysis mentioned in the first paragraph?

- ☐ a) Individual behaviour may depend on the resources available to attain certain ends.
- ☒ b) Individual behaviour is adversely impacted by scarcity of resources. **✗ Your answer is incorrect**
- ☐ c) Individual behaviour influences the ends chosen based on the resources.
- ☐ d) Individual behaviour is a function of availability of resources.

Q3. Which of the following, if true, weakens the Paretian basis of the anti-hedonistic argument mentioned in the first paragraph of the passage?

- ☐ a) It is not possible to objectively compare incentives across various possibilities.
- ☒ b) **There could be multiple possibilities offering the same incentive.** ✖ **Your answer is incorrect**
- ☐ c) **There could be a single possibility that offers multiple incentives in different situations.**
- ☐ d) The incentives offered by various possibilities are not influenced by the scarcity of those possibilities.

Q4. The author dubs the means-ends rationality 'psychical not physical' because:

- ☐ a) There is no data to prove that an individual's subjective valuation is rational.
- ☒ b) **There is no way to objectively measure the valuation process of an individual.** ✓ **Your answer is correct**
- ☐ c) **Psychology is the only way of understanding facts of experience.**
- ☐ d) There is no physical evidence proving that a relationship between means and ends exists.

DIRECTIONS *for questions 5 to 8:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

The focus on Western military effectiveness distracts attention from an issue that is frequently overlooked if a military approach to imperialism is taken, but one that is nevertheless important when considering all empires. This is the issue of the extent to which the role of the outside, apparently conquering, power has to be understood in terms of the power-plays of local interests, especially allies, so that the conqueror is used and, indeed, becomes effective only insofar as it takes part in such relationships. This reciprocal, patron-client relationship was critical to British control of India, while in southern Africa, Britain benefited from native opposition to the Zulus, whose expansion, like that of the Asantes and of Samory Toure´ in West Africa, had left much discontent.

The French also had some success with non-military efforts at pacification and assimilation, particularly in Senegal, Madagascar, and Morocco. The problems of pacifying large areas of often-difficult terrain ensured that co-operation was important. Thus, in Burma, where King Thebaw was defeated in the Third Anglo-Burmese War in 1885 and then surrendered, opposition continued, with a widespread rebellion in 1886–90. In confronting the situation there, the British benefited from the use of Indian military resources and also from some local support. Difficulties in pacification could also lead to a harshness that can be seen as an aspect of total war. For example, aside from refusing to take prisoners and killing the wounded—both seen in British operations against the Zulus—there was also the use of destruction, such as the burning down of the Asante capital Kumasi in 1874, and the devastation of Maori villages in New Zealand, in order to end resistance via humiliation and a statement of power.

Such methods, as well as the ability to win local support, can be seen as particularly important if the distinction between battle and war is grasped. The range of factors that led to Western victory in battle – including, as the nineteenth century drew toward its close, more deadly weapons, such as machine guns and effective cannon – did not necessarily ensure success in war or pacification. Instead, relatively small Western forces sought in their own apparent interest to direct states and peoples, which often meant claiming formal sovereignty, but found that the means available for doing so were limited. As a consequence, they took part in local power-plays, particularly recruiting local military support, which was used with considerable success by all imperial powers.

Native military support continued to be the pattern for Western imperialism in the twentieth century, whether on the offensive or on the defensive. In December 1934, an Italian force, composed largely of Somali irregulars, played the key role in a border clash at Walwal, which was used by the Italian dictator Benito Mussolini to justify the conquest of Ethiopia in 1935–36. In this conquest, the Italians benefited not only from Eritrean auxiliaries, but also from native allies such as the Oromo, whom they armed. An emphasis on the role of local power-plays risks demilitarizing military history, but this perspective is important: battle was an enabler of war, and war of the extension of control, but neither exhausts the subject, nor provides a sufficient explanation of success.

Q5. Which of the following, if true, provides the least support to the author’s arguments in the passage?

- ☐ a) In 1901, the British force that repelled the army of Mullah Sayyid Muhammed, when the latter attacked its camp in Somalia, was largely composed of Somalis.
- ☐ b) **Japan benefited in its conflict with China in 1894–95 because it was more united, not least in contrast with the acute divisions between the Manchu rulers of China and their Chinese subjects.**
- ☐ c) **The British force sent against Kano in what became northern Nigeria in 1902 was essentially one of White officers and eight hundred Nigerian troops.**
- ☐ d) In 1887-88, King Tieba of Ke’ne’dougou sought French aid when attacked by Samory Toure´ in modern Mali of Africa.

Q6. The example of an Italian force composed largely of Somali irregulars used by the Italian dictator Benito Mussolini (fourth para) was provided by the author to argue that

- ☐ a) the conquest of Ethiopia was done under false pretext.
- ☐ **b) Italians benefited more from the Ethiopian conquests than did their local allies.**
- ☐ **c) local power-play plays a greater role than an actual military in some conquests.**
- ☐ d) Western imperialism thrived through support from local military support.

Q7. From the statement “An emphasis on the role of local power-plays risks demilitarizing military history, but this perspective is important...”, the author probably means that

- ☐ a) there is more to a war than just military exploits in a battle.
- ☐ **b) military history should include the role of local power-plays for a true narrative of war.**
- ☐ c) **wars are more about who has control than about who is more militarily effective.**
- ☐ d) battles are about military but wars are about control.

Q8. The purpose of the first para is to

- ☐ a) highlight the importance of Western military effectiveness in aiding its imperialistic expansion.
- ☐ **b) show how the British benefited in their imperial ambitions from local power-plays.**
- ☐ c) argue that the role of outside powers in local powerplays is critical to understanding imperialism.
- ☐ d) prove that imperial expansion was only possible because of the powerplays around local interests.

DIRECTIONS *for questions 9 to 12:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Schopenhauer's arguments anticipate by over a century some of those offered by Peter Strawson in his important book, *Individuals*. But there is an important difference between Schopenhauer's approach and that of Kant, and of neo-Kantians like Strawson. Kant and Strawson primarily investigate the question of the basic or "a priori" structures of experience needed for a coherent cognitive life. But Schopenhauer, in seeking to explain why we are willing to use these structures even though they inevitably conceal reality from us, does not limit himself to exclusively cognitive considerations. He considers these structures to be primarily enablers of action rather than of knowledge. This shift should not be surprising. It is natural to view the purportedly universal activity of structuring and distorting our own experience as serving our fundamental needs. And what one takes these needs to be depends upon one's view of the fundamental nature of human beings. Schopenhauer, who identified our fundamental nature as will, suggested that we impose distorting structures upon experience - and thereby create a world of identifiable particulars - not primarily because this is a necessary condition of being able to comprehend the world rationally, but rather because only such distortion enables us to act.

Schopenhauer's line of argumentation, unpacked and somewhat elaborated, would be something like this: First, human beings need to be able to be conscious of particular individual things not primarily in order to have knowable objects, but rather to have determinate objectives for their actions. The objectives of action, the states of affairs and events that we seek to bring about, must be constituted by individual things; for without individuals there would be nothing that could serve as an objective of action, as an objective toward which the will could strive. And without objectives there could be no action...

Second, this framework allows each one of us to appear as an individual body in a world of individuals, and thus as the necessary locus of agency, the center from which desires and action can originate. Third, without the efficacy of action that causality brings to the world, it would be impossible to undertake to do anything, to accomplish a task or act upon the world. What would it mean even to attempt to bring something about in a world in which one's efforts had no efficacy, that is, in which they were not causally connected with the realization of the states or events desired? According to this account, the fundamental structures of our experience primarily serve the most basic need of the will, which is to be able actively to will something. Since Schopenhauer holds that "will" is what we really are, the essence of our true being, the most basic need of the will is our most basic need. Thus, it becomes plausible that we would be ready to subordinate and even sacrifice all other human needs and desires to its requirements for active agency. Schopenhauer's account of why we veil the world from ourselves, in its appeal to the idea of noncognitive interests that might lie deeper than our cognitive concerns and override them, opened the way for other variants of this strategy. Nietzsche's account is one such variant.

Q9. Which of the following was not mentioned as one of the salient features of Schopenhauer's line of argumentation?

- ☐ a) To identify an objective that one can strive towards
- ☐ b) **To identify oneself as an agent of change**
- ☐ c) **To bring about actual change through action**
- ☐ d) To identify our fundamental nature through cognitive considerations

Q10. The author mentions that 'it becomes plausible that we would be ready to subordinate and even sacrifice all other human needs and desires to its requirements for active agency,' to conclude that:

- ☐ a) Our requirement for active agency is our most basic need.
- ☐ b) **The most basic need of the will is our most basic need.**
- ☐ c) **We need to veil ourselves from the world.**
- ☐ d) Human needs and desires precede active agency.

Q11. Which of the following is incompatible with Schopenhauer's argumentation?

- ☐ a) Our will to act supersedes our needs and desires.
- ☐ b) **Our need to understand the world is secondary to the needs of our will.**
- ☐ c) **Our will is overridden by our needs and desires.**
- ☐ d) The will to act becomes our fundamental need.

Q12. When the author says that 'this shift should not be surprising', the author means to assert that:

- ☐ a) A priori structures of experience enable action more than knowledge according to Schopenhauer.
- ☐ b) **Our fundamental needs are not served by distorting our knowledge and experience.**
- ☐ c) **A priori structures of experience are needed for coherent cognition.**
- ☐ d) A priori structures of experience conceal reality from us.

DIRECTIONS *for questions 13 to 16:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Unitarity, as the principle is called, says that something always happens. When particles interact, the probability of all possible outcomes must sum to 100%... But what once seemed an essential scaffold may have become a stifling straitjacket preventing physicists from reconciling quantum mechanics and gravity. A century ago, the pioneers of quantum mechanics made a surprising discovery – one that elevated unitarity from common sense to a hallowed principle. Mathematically, the quantum world operates by more complicated numbers known as amplitudes. An amplitude is essentially the degree to which a particle is in a certain state; it can be a positive, negative or imaginary number. To calculate the probability of actually observing a particle in a certain state, physicists square the amplitude which gets rid of the imaginary and negative bits and produces a positive probability. Unitarity says the sum of these probabilities must equal 1... As a particle's state changes, its amplitudes change too. In working out how a particle is allowed to evolve or interact, physicists use the fact that amplitudes never change in a way that disrupts the fixed sum of their squares.

Physicists keep probabilities and amplitudes in line by tracking how the quantum state of a particle moves around in Hilbert space – an abstract space representing all possible states available to the particle. The particle's amplitudes correspond to its coordinates in Hilbert space... Unitarity dictates that a physically allowed change must correspond to a special “unitary” matrix that rotates the particle's state in Hilbert space such that the sum of the squares of its coordinates equals 1.

And yet, this bedrock assumption seems to conflict with the universe that surrounds us. While our expanding universe is a perfectly valid solution to the equations of general relativity, physicists have increasingly realized that its growth spells trouble for quantum mechanics, by presenting particles with an expanding smorgasbord of options for where to be and how to behave. As space grows, how can the Hilbert space of possibilities not grow with it?

Andrew Strominger and Jordan Cotler of Harvard University argue that a more relaxed principle called isometry can accommodate an expanding universe while still satisfying the stringent requirements [of] unitarity. In a paper earlier this year, the two homed in on a class of transformations known as isometries. An isometric change resembles a unitary one with added flexibility. Think of an electron that can occupy two possible locations. Its Hilbert space consists of all possible combinations of amplitudes in the two locations. These possibilities can be imagined as the points on a circle — every point has some value in both the horizontal and vertical directions. Unitary changes rotate states around the circle but do not expand or shrink the set of possibilities.

To visualize an isometric change, though, let the universe of this electron swell just enough to allow a third position. The electron's Hilbert space grows, but in a special way: It gains another dimension. The circle becomes a sphere, on which the particle's quantum state can swivel around to accommodate mixtures of all three locations. The distance between any two states on the circle holds steady under the change — another requirement of unitarity. In short, the options increase, but without unphysical consequences...

Q13. The term ‘stifling straitjacket’ taken in the context of the passage is most closely associated with which of the following constraints?

- ☐ a) The Hilbert space of all possible available states is hard to trace for every particle such that it satisfies unitarity.
- ☐ b) **The number of possible states in which a particle should be present in an expanding universe is quite limiting.**
- ☐ c) **The possibilities for particle states in an expanding universe may not be confined to the earlier version of Hilbert space.**
- ☐ d) Probabilities and amplitudes have to be kept in line by tracking how the quantum state of a particle moves around in Hilbert Space.

Q14. Which of the following best represents the suggestion put forth by Strominger and Cotler in order to reconcile the Unitarity principle with the idea of an expanding universe?

- ☐ a) Expand the quantum states of an electron to three instead of two
- ☐ b) **Use a three-dimensional figure for capturing the possible states available to a particle instead of a two-dimensional one**
- ☐ c) **Represent all possible states available to a particle using more dimensions**
- ☐ d) Enhance the probabilities of an electron's presence in three locations

Q15. Which of the following statements will the author least approve of?

- ☐ a) The sum of all probabilities of observing a particle in various states is 1.
- ☐ b) **Amplitudes of particles are squares to avoid imaginary and negative components.**
- ☐ c) **Unitarity helps in ensuring that all probabilities related to a particle's state are accounted for.**
- ☐ d) The degree to which a particle is in a certain state controls its physical changes.

Q16. The author's statement that 'this bedrock assumption seems to conflict with the universe that surrounds us' is based on the presupposition that:

- ☐ a) An expanding universe offers more space for particles to be in more possible states.
- ☐ **b) An expanding universe is far more chaotic than what we supposed it to be.**
- ☐ c) **An expanding universe will have a greater number of particles than previously assumed.**
- ☐ d) An expanding universe is perfectly compatible with general relativity.

Q17. DIRECTIONS *for questions 17 and 18:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

There is certainly something paradoxical about the idea of a logic whose terms consist of odds and ends left over from psychological or historical processes and are, like these, devoid of necessity. Logic consists in the establishment of necessary connections and how, we may ask, could such relations be established between terms in no way designed to fulfil this function. Propositions cannot be rigorously connected unless the terms they contain have first been unequivocally defined.

- ☐ a) Logical propositions cannot be derived from ill-defined psychological or historical processes.
- ☐ b) **It is not possible to connect psychological or historical processes to a logically defined function.**
- ☐ c) **Logical propositions become paradoxical when one makes connections out of psychological and historical processes.**
- ☐ d) A logical proposition cannot be constructed from ill-defined terms that do not assist in making connections.

Q18. DIRECTIONS *for questions 17 and 18:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

Consciousness is, by definition, subjective, in the sense that for a conscious state to exist it must be experienced by some conscious subject. Consciousness in this sense has a first-person ontology in that it only exists from the point of view of a human or animal subject, an “I,” who has the conscious experience. Science is not used to dealing with phenomena that have a first-person ontology. By tradition, science deals with phenomena that are “objective,” and avoids anything that is “subjective.”

- ☐ a) Science cannot define consciousness because of the subjective nature of the latter.
- ☐ b) **Science and consciousness are fundamentally in opposition to each other.**
- ☐ c) **Science and consciousness cannot be defined together because one is subjective and the other isn't.**
- ☐ d) Science's objectivity makes it ill-equipped to explain the subjective phenomenon of consciousness.

Q19. DIRECTIONS *for questions 19 and 20:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Topology allows mathematicians to look past minor variations to focus on fundamental ways in which different shapes are the same.
2. Higher symmetries can detect that — and by detecting it, they allow physicists to take knowledge about better-understood quantum systems and apply it to others.
3. Those systems may look completely different from each other, but in a deep way they might really obey the same rules.
4. Similarly, higher symmetries provide physicists with a new way of indexing quantum systems.

Q20. DIRECTIONS *for questions 19 and 20:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Some historians maintained that operational art first emerged in the American Civil War and Moltke’s campaigns in 1866 and 1870–1871 as a child of the Industrial Revolution.
2. The term ‘operational art’, coined by Soviet military theorists in the interwar period, has received increased attention in military circles with the debate on comparative NATO and Warsaw Pact operational capabilities during the 1970s and 1980s.
3. Proponents of this position hold that the invention of rifled guns and infantry arms, the railway and the telegraph permitted the widely dispersed manoeuvre of independent bodies of troops which is a salient feature of operational art.
4. The interest in operational art also extended into the field of military history when the origins of operational art became the object of research.

Q21. DIRECTIONS *for question 21:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: In fact, they take positivity to some shocking extremes.

Paragraph: Linkin Park could have been designed in a laboratory as the consummate rap-metal band, circa 2001: the rage-filled vocals, the headbanging guitar, the renegades-of-funk rhythm section, the DJ scratching between verses, the shy, intense guitarist with the arty tastes. ____ (1) _____. But although their heavy, aggressive sound gets them lumped in with bands like Korn and Limp Bizkit, Linkin Park are hardly the party commandos or death freaks you might expect. Instead, they're sensitive dudes who sing about the secret life of boys: the real-life emotional struggles of ordinary guys like themselves, hitting a nerve in the audience with their brotherly compassion. ____ (2) _____. Like Papa Roach and Incubus, Linkin Park take pride in keeping it positive. ____ (3) _____. For all the rage and fury in the music, they feel your pain. ____ (4) _____. None of the band members utters a single curse word on *Hybrid Theory*.

- ☐ a) Option 1
- ☐ b) **Option 2**
- ☒ c) **Option 3** ✖ **Your answer is incorrect**
- ☐ d) Option 4

Q22. DIRECTIONS *for question 22:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

Despite the obvious interest aroused by battle, murder and sudden death, the Wars of the Roses have been astonishingly neglected by modern historians, and even by the more recent historical novelists. Medieval History ebbs with the Hundred Years War, and Modern History begins with the Tudors, while, between the two, the tide of interest leaves the wars of Lancaster and York high and rather dry. The growing concern with social history has merely accentuated this neglect: the Wars of the Roses may be mentioned incidentally in a chapter dealing with the general lawlessness of the times, but they are regarded principally as a matter for the political or military historian, to whom, on the other hand, they often seem little more than a “glorified tournament.”

- ☐ a) The Wars of the Roses have been relegated to the backseat in history because of the popularity of the Hundred Years War and the history of the Tudors.
- ☐ b) **Despite involving elements of interest, historians have ignored Wars of the Roses from accounts of Medieval and Modern History.**
- ☐ c) **Historians have wrongly neglected the Wars of the Roses in their accounts of Medieval History.**
- ☐ d) The Wars of the Roses doesn't find space in Medieval History because of the lack of interest from political and military historians.

Q23. DIRECTIONS *for question 23:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Safety, Winston Churchill remarked in one of his last speeches as prime minister, had become 'the sturdy child of terror.'
2. However, few of those responsible for maintaining the peace were content to rely on terror alone.
3. The nuclear age began with the Cold War, some would say simultaneously, for as soon as the United States revealed the unique power at its command, the Soviet Union began to make immediate adjustments to its own behaviour.
4. The dread of nuclear exchanges provides one explanation for how the world survived such deep superpower antagonism without another total war.

Q24. DIRECTIONS *for question 24:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: Today, the Nike swoosh is so ubiquitous, it’s hard to remember a time when the athletic-wear giant was the underdog in the basketball shoe business.

Paragraph: _____(1)_____. A decade after “Argo” won best picture, Ben Affleck brings his populist instinct – plus, all the competitive energy of a classic sports movie – to the boardroom hustle it took for Nike to convince a rookie named Michael Jordan to lend his name to the company’s fledgling ‘Air’ line. _____(2)_____. Like “Moneyball” and “Jerry Maguire,” the movie promises to reveal hidden dimensions of the sports industry without losing track of the human element. _____(3)_____. Here we cheer as a corporate suit (Matt Damon, running on faith and flop sweat) pitches Phil Knight (Affleck) first, and then the one person who matters: Jordan’s mom (Viola Davis). _____(4)_____.

- ☐ a) Option 1
- ☐ b) **Option 2**
- ☐ c) **Option 3**
- ☐ d) Option 4